

1. AIM

To design and verify the following logic gates using domino logic and calculate their performance parameters such as threshold voltage, noise margin, propagation delay, rise time, and fall time: 1) Dynamic CMOS Domino Inverter, 2) 2-input NAND Gate.

2. TOOL

Cadence Virtuoso

3. THEORY

Domino logic is a dynamic CMOS logic family that uses a precharge phase and an evaluation phase to perform logic operations. It consists of a PMOS precharge transistor, an NMOS evaluation network, and a static CMOS inverter at the output. During the precharge phase, the dynamic node is charged to V_{DD} , and during the evaluation phase, the NMOS network conditionally discharges the dynamic node according to the input logic.

A dynamic CMOS domino inverter consists of an NMOS transistor in the evaluation path controlled by the input signal, followed by a static CMOS inverter. When the input is low, the dynamic node remains charged and the output is low. When the input is high during evaluation, the dynamic node discharges and the output becomes high, producing the required inversion.

A 2-input NAND gate is implemented using an NMOS evaluation network with two NMOS transistors connected in parallel. During evaluation, the dynamic node discharges when either input is high, and the following static inverter produces the NAND logic function at the output.

The performance of the designed circuits is evaluated using their voltage transfer characteristics and transient response. The threshold voltage and noise margin are obtained from the voltage transfer characteristics, while propagation delay, rise time, and fall time are determined from the transient response.

The circuits are designed and verified using Cadence Virtuoso, and the obtained performance parameters are analyzed to evaluate the operation of the domino logic circuits.

4. TABLES

Table 4.1: Design Parameters

Parameter	Value	Unit
Technology	180	nm
V_{dd}	1.8	V
PMOS Width	400	nm
NMOS Width	400	nm

Table 4.2: Performance parameters of Dynamic CMOS Domino Inverter

parameter	value of inverter	value of 2 input nand	unit
vth	718.473	731.301	mV
voh	1.8	1.8	V
vol	0	0	V
vih	836.611	836.613	mV
vil	685.147	685.527	mV
nmh	963.389	963.387	mV
nml	685.147	685.527	mV
tphl	0.0883	0.1073	V
tplh	0	0	V
tp	0.0883	0.1073	v
tr	0.0756	0.0759	V
tf	0.0315	0.0314	V

5. CALCULATION FOR TRANSMISSION GATE BASED D FLIP FLOP

For the Dynamic CMOS Domino Inverter:

$$V_{TH} = 718.473mV$$

$$NM_H = V_{OH} - V_{IH} = 1800 - 836.611 = 963.389mV$$

$$NM_L = V_{IL} - V_{OL} = 685.147 - 0 = 685.147mV$$

$$t_{PHL} = 31.2133 - 31.125 = 0.0883ns = 88.3ps$$

$$t_{PLH} = 0ns$$

$$t_r = 33.2219 - 33.1463 = 0.0756ns = 75.6ps$$

$$t_f = 31.2306 - 31.1991 = 0.0315ns = 31.5ps$$

For the 2-input NAND Gate:

$$V_{TH} = 731.301mV$$

$$NM_H = V_{OH} - V_{IH} = 1800 - 836.613 = 963.387mV$$

$$NM_L = V_{IL} - V_{OL} = 685.527 - 0 = 685.527mV$$

$$t_{PHL} = 31.1098 - 31.0025 = 0.1073ns = 107.3ps$$

$$t_{PLH} = 0ns$$

$$t_r = 32.5935 - 32.5176 = 0.0759ns = 75.9ps$$

$$t_f = 31.1271 - 31.0957 = 0.0314ns = 31.4ps$$

6. CIRCUIT DIAGRAMS & WAVEFORMS

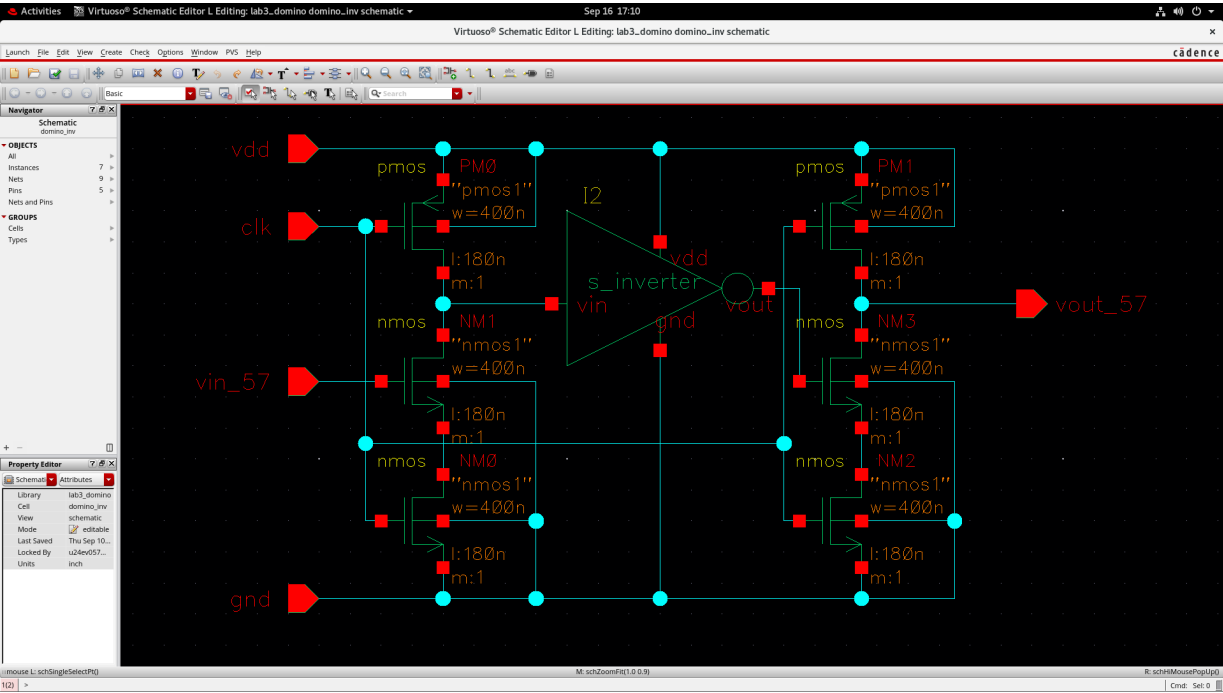


Figure 6.1: Inv Schematic

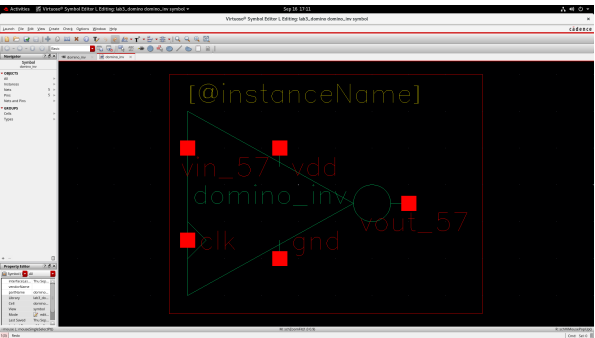


Figure 6.2: Inv Symbol

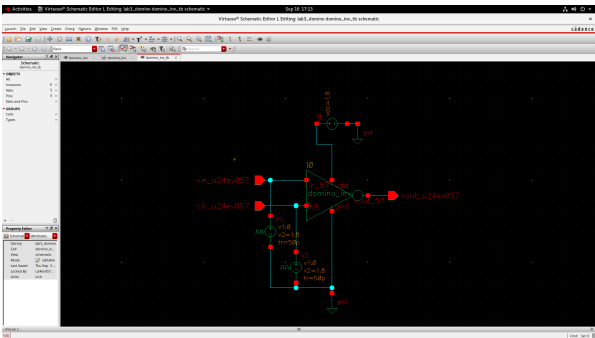
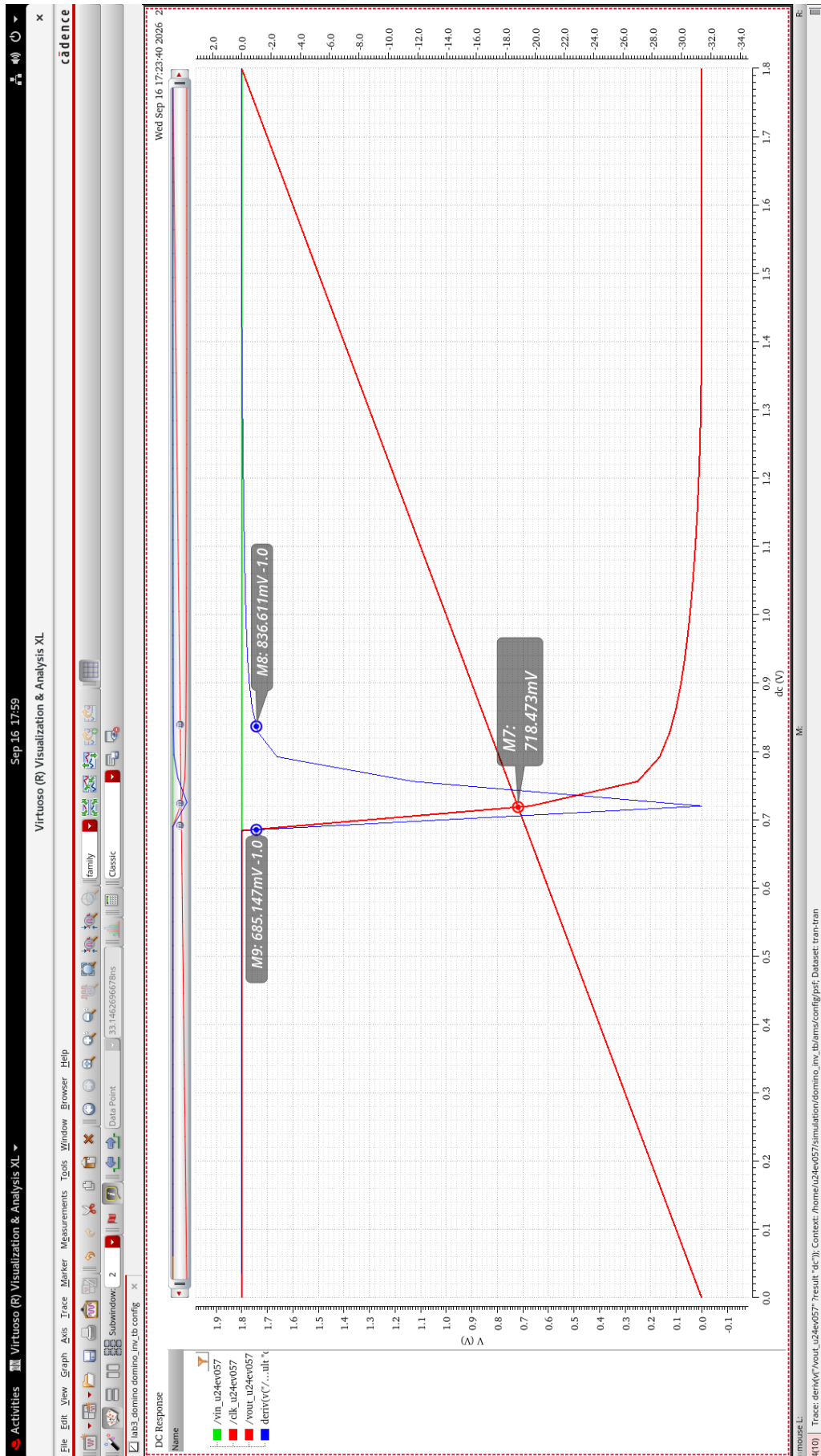


Figure 6.3: Inv Tb



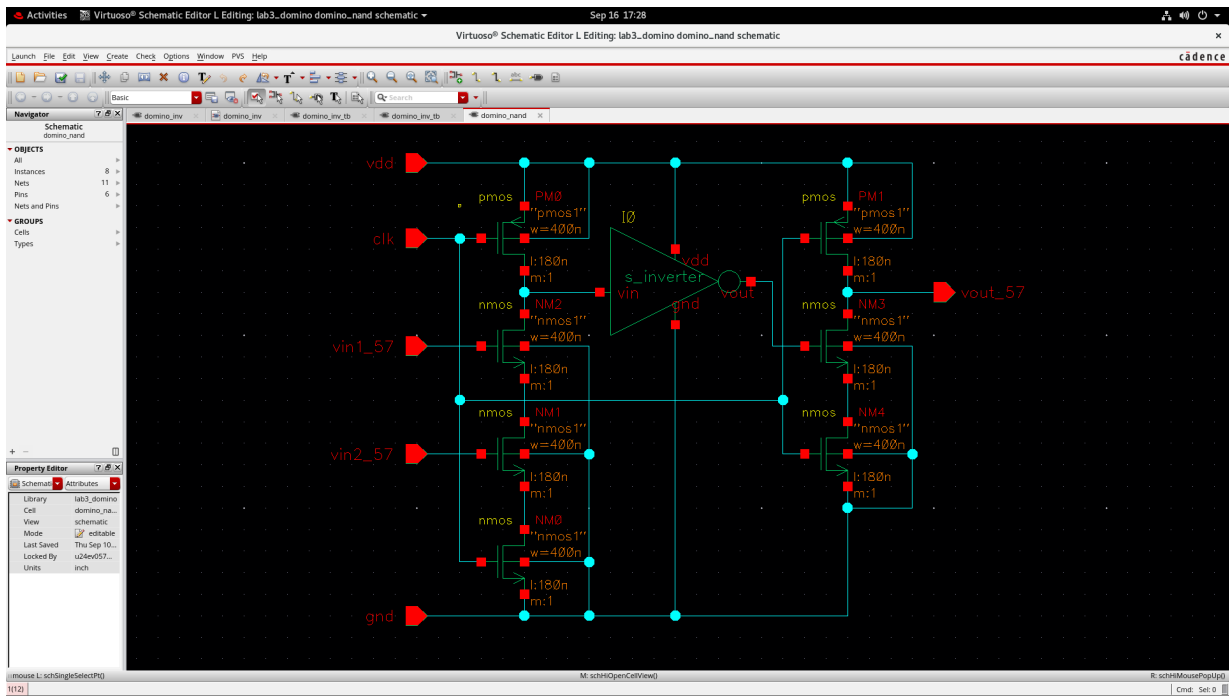


Figure 6.6: Nand Schematic

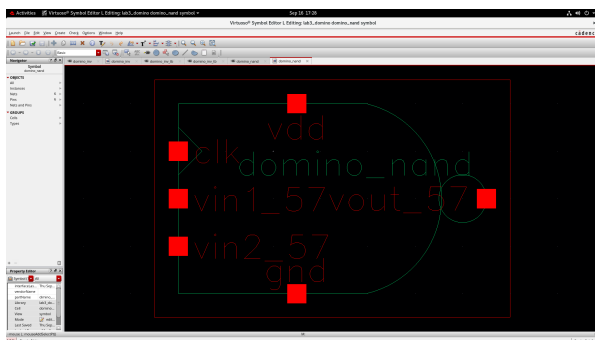


Figure 6.7: Nand Symbol

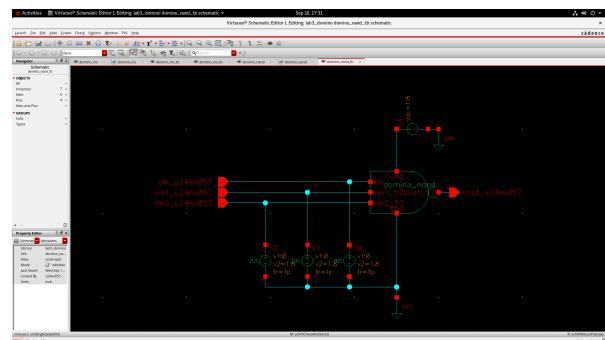


Figure 6.8: Nand Tb

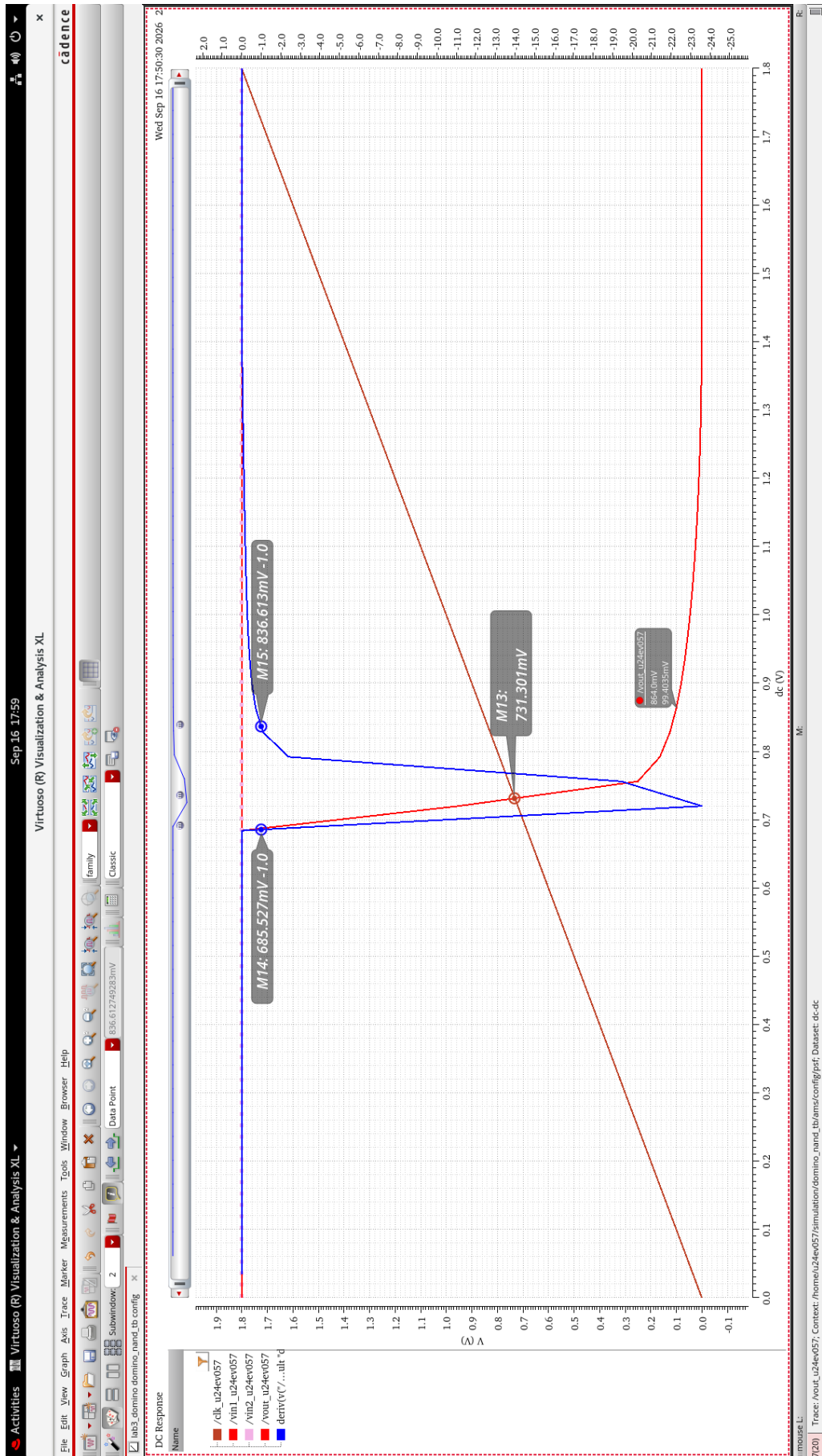


Figure 6.9: Nand Dc

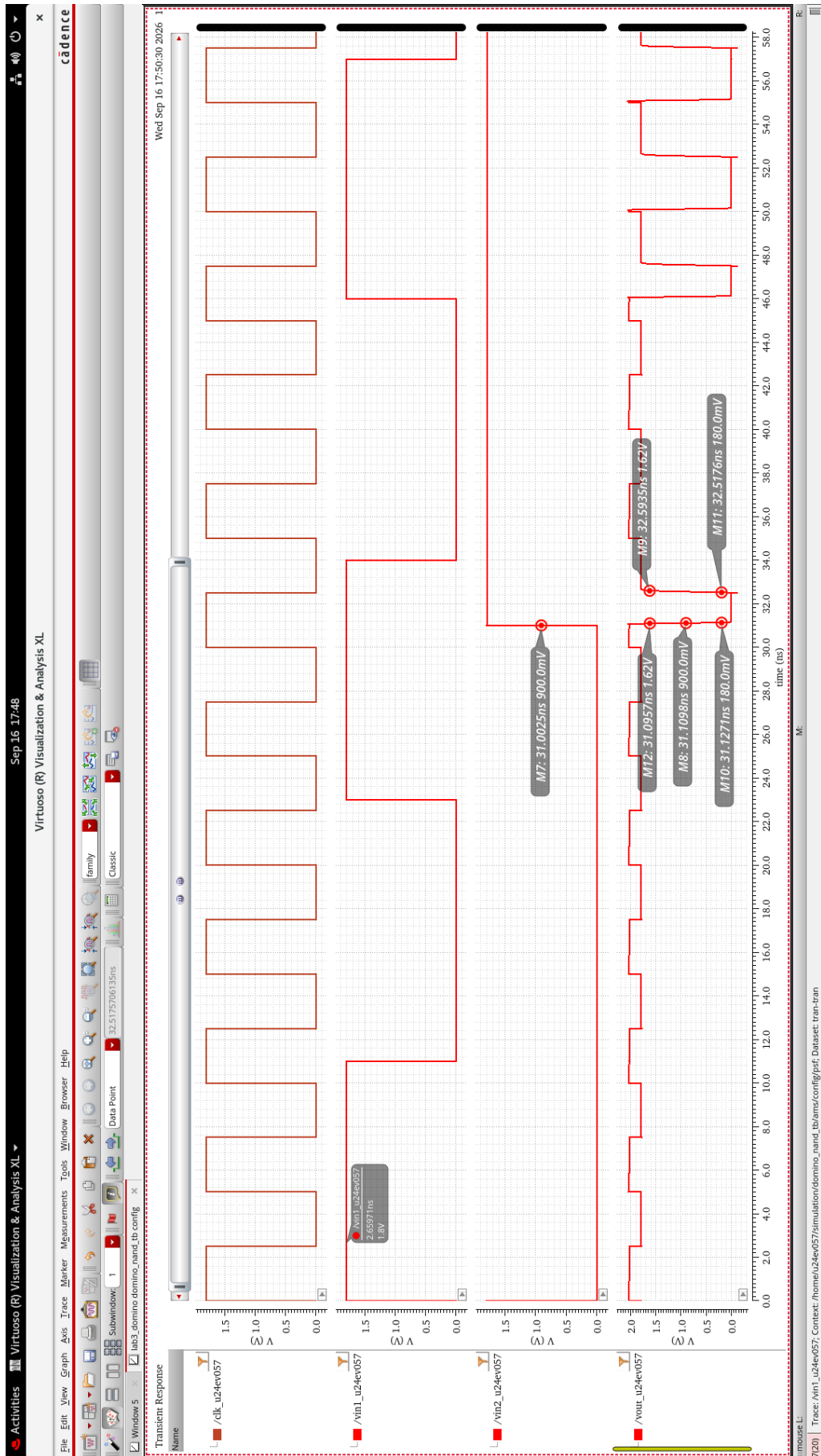


Figure 6.10: Nand Tran

7. RESULT

The Dynamic CMOS Domino Inverter and 2-input NAND Gate were successfully designed and verified. The threshold voltage, noise margins, propagation delay, rise time, and fall time were calculated from the simulated results, confirming the operation and performance of the designed domino logic circuits.